Indonesia NCD Simulation Model

Technical Overview — CVD and Cancer Markov Frameworks

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Part 1 — Project Overview

Background and Motivation

- Indonesia faces a growing NCD burden driven by population aging, high smoking prevalence, rising obesity, and limited screening access
- CVD and cancer together account for over 50% of premature mortality in Indonesia
- Policy decisions require quantitative evidence on intervention scale-up trajectories and their expected health and economic returns

Why Markov?

Unlike system dynamics approaches (e.g., NCDSim), discrete-time Markov models allow transparent, state-by-state tracking of disease progression—making calibration to GBD and validation against observed rates straightforward.

Aims and Scope

Primary aim: Project health and economic outcomes of NCD policy packages in Indonesia from 2025 to 2050.

Two independent disease modules:

1. **CVD** — IHD, ischemic stroke, hemorrhagic stroke, hypertensive heart disease
2. **Cancer** — 16 types from GBD 2021, in three analytic tiers

Policy levers modeled:

- Hypertension control scale-up (HEARTS)
- Sodium reduction
- Trans-fat elimination
- Statin coverage expansion
- Cancer screening (cervical, colorectal, breast)

Reproducible Research Workflow

Inputs $\longrightarrow$ Scenario Definitions $\longrightarrow$ Model Outputs $\longrightarrow$ Documentation $\longrightarrow$ Reports

Stage	Source
Raw data	GBD 2023, UNWPP 2024, IHME, NCD-RisC, ETIHAD
Scenarios	`04_define_interventions.R` / `settings.yml`
Simulation	Markov engine per module
Post-processing	DALYs, VSL/VSLY calculations
Reporting	Beamer slides, R Markdown, policy briefs

Methods Note

All scenario definitions are code-driven and version-controlled—no manual spreadsheet steps in the analytic pipeline.

Part 2 — Technical Infrastructure

Agentic Coding Approach

- Development assisted by Claude Code (Anthropic) for code generation, review, and documentation
- AI-assisted workflow embedded in version control via `CLAUDE.md` project instructions
- Enables rapid iteration on intervention logic and scenario definitions while maintaining auditability

Key Insight

The agentic approach accelerates development of complex modules (e.g., 850+ line HTN control logic) while preserving human oversight through structured code review and calibration checks.

Reproducibility Infrastructure

What is currently in place:

- Organized directory layout: `data/raw`, `data/interim`, `data/processed`
- Configuration via `settings.yml` (cancer) and flag variables (CVD)
- Numbered script execution order (00–08 for CVD; 00–06 for cancer)
- Intermediate `.rds` checkpoints at each pipeline stage
- Remote function sourcing with local fallback paths

Not yet in place:

- No `renv.lock` — dependencies installed manually
- No formal seed control for stochastic components
- No containerized environment (Docker)

Reproducibility Gap

Adding `renv` lockfiles and a `Dockerfile` would close the main reproducibility gaps. These are planned but not yet implemented.

Version Control and GitHub Workflow

- Repository hosted on GitHub under `Disease-Control-Priorities`
- Branch-based development with pull request reviews
- `CLAUDE.md` serves as machine-readable project documentation, checked into version control
- Raw data files excluded via `.gitignore`; processed `.rds` files tracked selectively

Data Governance

Raw GBD/UNWPP data are not stored in the repository. Analysts must obtain them independently and place them in `data/raw/`.

Part 3 — Methodological Overview

Full Model Architecture

Population → Risk Factors → Disease Models → Outcomes

Component	Status	Key Data Source
Demography / Population	Scaffolding only	UNWPP 2024
Risk factor distributions	Implemented (CVD)	NCD-RisC, IHME DIET
CVD natural history	Fully implemented	GBD 2023
Cancer natural history	Fully implemented	GBD 2021/2023
Health systems / treatment	Partial (coverage params)	ETIHAD, HEARTS
Economic evaluation	Fully implemented	Robinson VSL, World Bank GNI
Module integration	Separate pipelines	—

Key Insight

CVD and cancer modules run independently under a conditional independence assumption. No comorbidity interaction is modeled in v1.0.

Core Model Logic: CVD

State space (per age / sex / location / cause / year):

$$\text{Susceptible} \rightarrow \text{Incident} \rightarrow \text{Prevalent} \rightarrow \text{Dead}$$

- Transition probabilities calibrated to GBD 2023 incidence, prevalence, and mortality
- Background mortality layered from UNWPP 2024 life tables
- Secular trends in case fatality forecast via ARIMA (`forecast` package)

Methods Note

Constraint enforced: $IR + BG_{mx} \leq 1$ for all age–sex–cause–year cells. Violations are flagged in `032_adjustments.R` and corrected before simulation.

Core Model Logic: Cancer

7-state Markov model (per age / sex / location / cause / year):

$$\text{Well} \longrightarrow \text{Precancer} \longrightarrow \text{Local} \longrightarrow \text{Regional} \longrightarrow \text{Distant} \longrightarrow \text{Cancer Death}$$

with absorbing background death state.

Three analytic tiers:

1. **Tier 1** (11 cancers) — Incidence/mortality only (no staging)
2. **Tier 2** (3 cancers) — Staged: local, regional, distant
3. **Tier 3** (2 cancers) — Precancerous lesions modeled (cervix, CRC)

Methods Note

Tier 3 cancers include screening interventions that shift stage distribution at diagnosis, directly affecting transition probabilities.

Intervention Logic

CVD interventions are applied multiplicatively to incidence rates:

$$IR_{\text{treated}} = IR_{\text{baseline}} \times (1 - \text{coverage} \times RR_{\text{effect}})$$

Multi-intervention stacking order:

1. Sodium $\rightarrow$ BP distribution shift via `get.bp.prob()` coefficients
2. Statins $\rightarrow$ direct IR reduction (RR from meta-analyses)
3. TFA elimination $\rightarrow$ IR reduction via $RR = 1.28$ per 2% energy from TFA
4. HTN control $\rightarrow$ ETIHAD-based RR per 10 mmHg BP reduction

Cancer interventions: Screening coverage expansion shifts stage distribution at diagnosis, reducing late-stage presentation and improving survival.

Economic Evaluation Framework

Value of Statistical Life (VSL) approach (Robinson et al., 2019):

$$VSL_{\text{country}} = VSL_{\text{base}} \times \left(\frac{GNI_{\text{country}}}{GNI_{\text{base}}}\right)^{\varepsilon}$$

- Three income elasticity scenarios: $\varepsilon = 1.0, 1.2, 1.5$ (primary: 1.2)
- VSLY derived: $VSLY = VSL/\bar{e}$ (average adult life expectancy)
- GDP growth projections from SSP2 scenario

Takeaway

Economic valuation converts deaths averted and DALYs prevented into monetary terms, enabling comparison across intervention packages and informing resource allocation decisions.

How Modules Integrate

Current state: CVD and cancer pipelines run independently, producing separate outputs (deaths averted, DALYs, economic value).

Integration points:

- Shared demographic inputs (UNWPP 2024 population, life tables)
- Shared economic framework (VSL/VSLY methodology)
- Common reporting templates (Beamer, R Markdown)

Not yet integrated:

- No joint population projection feeding both models
- No comorbidity or competing-risk adjustment across CVD and cancer
- No unified scenario configuration file

Integration Pending

A formal integration layer—shared population module, unified scenario config, and joint output aggregation—is planned for v2.0.

Part 4 — CVD Module Walkthrough

CVD Pipeline Architecture

```
00_run_model.R              (orchestrator + config flags)
  |-- 01_utils.R            (BP control, TFA policy, age grouping)
  |-- 02_load_inputs.R   (sources 021--023)
  |     |-- 021_get_base_rates.R   (GBD 2023 extraction)
  |     |-- 022_get_tps.R          (transition probabilities)
  |     |-- 023_get_tps_bgmx.R     (background mortality trends)
  |-- 03_calibration.R   (sources 031--032)
  |     |-- 031_calibration.R      (calibrate to GBD)
  |     |-- 032_adjustments.R      (IR + BG.mx <= 1)
  |-- 04_define_interventions.R  (scenarios, ~57 KB)
  |-- 05_build_baseline.R          (cohort assembly)
  |-- 06_run_scenarios.R           (simulation engine, ~84 KB)
  |-- 07_output_dalys.R            (YLL, YLD, DALYs)
  |-- 08_economic_value_calculation.R  (VSL/VSLY)
```

Key Insight

Configuration flags in `00_run_model.R` allow skipping expensive stages when intermediate `.rds` files already exist.

CVD: Data Loading and Calibration

Data sources ingested:

- GBD 2023: cause-specific mortality, incidence, prevalence (4 CVD causes)
- UNWPP 2024: population by age/sex, life expectancy, fertility
- NCD-RisC: historical blood pressure distributions (1990–2019)
- IHME DIET: sodium intake estimates

Calibration pipeline:

1. Extract single-year age rates (20–95) from 5-year GBD groups via interpolation
2. Compute transition probabilities from rate-to-probability conversions
3. Enforce $IR + BG_{mx} \leq 1$ constraint; flag and correct violations
4. Forecast secular trends in case fatality using ARIMA

Methods Note

Inline validation: `print(anyNA(data.out))` after every merge/join in `021_get_base_rates.R`.

CVD: Intervention Definitions

Four intervention families defined in `04_define_interventions.R`:

Intervention	Scale-up	Target	Key Parameter
Sodium	Linear 2025–2030	-15% / -30%	BP shift coefficients
TFA	Linear to 0%	Elimination	$RR = 1.28$ per 2% TFA
Statins	Linear / logistic to 2050	60% coverage	LDL meta-analysis RRs
HTN control	Quadratic (NCD-RisC fit)	3 scenarios	ETIHAD RRs per 10 mmHg

HTN control is the most complex module (~850 lines), fitting quadratic scale-up functions to historical NCD-RisC data and producing coverage trajectories (Business-as-usual, Progress, Aspirational).

Takeaway

Output: `covfxn2.csv` with columns for location, year, coverage scenario, absolute change, and annual rate of change.

CVD: Simulation Engine

`06_run_scenarios.R` (~84 KB) is the primary simulation engine:

- Iterates over all age/sex/location/cause/year combinations
- Applies intervention effects via coverage-adjusted relative risk reductions
- Supports parallel execution across intervention combinations
- Outputs: deaths averted, incident cases prevented, prevalent cases

Intervention stacking applies effects sequentially to avoid double-counting:

1. Sodium $\rightarrow$ shifts BP distribution (population-level)
2. Statins $\rightarrow$ reduces IR directly (individual-level)
3. TFA $\rightarrow$ reduces IR via RR (population-level)
4. HTN control $\rightarrow$ applies ETIHAD RR to treated fraction

Insert figure placeholder: *`output/slides/cvd_scenario_comparison.png` — Deaths averted by intervention package, 2025–2050*

CVD: DALY and Economic Outputs

Burden calculation (`07_output_dalys.R`):

- YLL = deaths $\times$ residual life expectancy (UNWPP 2024)
- YLD = prevalent cases $\times$ GBD disability weights (cause-specific)
- DALYs = YLL + YLD, aggregated by age, sex, cause, year

Economic valuation (`08_economic_value_calculation.R`, ~42 KB):

- VSL calibrated to Indonesia GNI with income elasticity $\varepsilon = 1.2$
- GDP growth from SSP2 projections (IIASA)
- Outputs: monetized value of deaths averted, DALYs prevented, cost per DALY averted

Insert figure placeholder: `output/slides/cvd_dalys_averted.png` — Cumulative DALYs averted by scenario

Insert figure placeholder: `output/slides/cvd_economic_value.png` — Economic value of intervention packages

Part 4 (cont.) — Cancer Module Walkthrough

Cancer Pipeline Architecture

```
library.R                  (setup, loads packages + fnx/)
  |
  |-- fnx/                 (23 reusable functions)
  |    |-- project_markov_trace_dt.R  (core projection)
  |    |-- calc_bsln_tps_dt.R         (baseline TPs)
  |    |-- calc_intv_tps.R            (intervention TPs)
  |    |-- calc_metric_dt.R           (GBD extraction)
  |    |-- run_ccpm.R                 (cohort component projection)
  |    |-- gen_lifetable.R            (abridged life tables)
  |    |-- process_intv_scen_inputs.R (scenario processing)
  |    +-- [16 other helpers]
  |
  |-- scripts/             (numbered execution)
       |-- 00_read_data.R
       |-- 01_process_country_param.R
       |-- 02_process_bsln_scen.R
       |-- 03_process_pop_proj_bsln_dt.R
       |-- 04_process_intv_scen.R
       |-- 05_process_pop_proj_intv_dt.R
```

Cancer: Three-Tier System

16 cancer types organized by analytic complexity:

Tier 1 — Incidence/mortality only (11 types): Uterine, esophageal, thyroid, stomach, liver, pancreatic, ovarian, bladder, lip/oral, nasopharynx, pharynx

Tier 2 — Staged (3 types): Lung, prostate, breast (local $\rightarrow$ regional $\rightarrow$ distant)

Tier 3 — Precancerous lesions modeled (2 types): Cervix, colorectal (well $\rightarrow$ precancer $\rightarrow$ local $\rightarrow$ regional $\rightarrow$ distant)

Key Insight

Tier 3 cancers are the primary targets for screening interventions, where early detection shifts stage distribution and reduces late-stage mortality.

Cancer: Core Markov Projection

`project_markov_trace_dt.R` implements the core engine:

- 7$×$7 transition probability matrix per (location, sex, cause, year, age)
- State vector: `pwell, pprc, plcl, prgn, pdst, pcdx, pbgdx`
- Forward projection via matrix multiplication, year-by-year
- Two absorbing states: cancer death, background death
- Handles newborn cohort initialization for population projection

Methods Note

Transition probabilities are first computed at calibration years, then expanded to the full projection period via `expand_tps_years_dt.R`. Matrix corrections enforced by `correct_markov.R`.

Cancer: Key Functions

Function	Role
`calc_bsln_tps_dt`	Baseline TPs from stage distributions + GBD rates
`calc_intv_tps`	Modify TPs for intervention scenarios
`calc_metric_dt`	Extract incidence/prevalence/mortality from GBD
`run_ccpm`	Cohort component population projection
`gen_lifetable`	Standard abridged life table (q_x, l_x, e_x, S_x)
`process_intv_scen_inputs`	Parse scenario YAML/Excel inputs
`frp_fnx`	Finite rate projection and RR adjustments
`calc_start_hstate_dist_dt`	Initial health state distribution
`id_target_cc_methods`	Identify cervical cancer screening methods

Cancer: Intervention Scenarios

Screening interventions processed by `process_intv_scen_inputs.R` (~17 KB):

- Define target coverage, efficacy, and ramp-up schedule
- Shift stage distribution at diagnosis (earlier detection)
- Modify transition probabilities accordingly via `calc_intv_tps`

Population projection via `run_ccpm.R`:

- Cohort component model: aging, births, deaths by age/sex
- Baseline and intervention projections run separately
- Difference yields intervention-attributable outcomes

Insert figure placeholder: `output/slides/cancer_stage_shift.png` — Stage distribution shift under screening scale-up

Takeaway

Cancer outputs mirror the CVD structure: incident cases, deaths, DALYs under base-line vs. intervention, enabling consistent cross-module comparison

Part 4 (cont.) — Pending Modules

Demography Module

Module Pending

The `code/demography/` directory exists as a placeholder. No implementation is present.

Planned scope:

- Unified population projection feeding both CVD and cancer modules
- Cohort construction with age/sex-specific mortality, fertility, migration
- Currently, each module handles demographics independently:
  - CVD: loads UNWPP data in `05_build_baseline.R`
  - Cancer: uses `run_ccpm.R` for cohort component projection

Integration benefit: A shared demography module would ensure consistent population denominators and avoid duplication.

Shared Utilities

Module Pending

The `code/utils/` directory exists as a placeholder. No shared utility functions are implemented.

Candidate functions for extraction:

- Rate $\leftrightarrow$ probability converters (currently sourced remotely in cancer module)
- Age-group harmonization (5-year $\leftrightarrow$ single-year)
- GBD data loaders and standardizers
- Common plotting themes and output formatters

Health Systems Module

Module Pending

No dedicated health systems module exists. Treatment and coverage parameters are embedded in intervention definitions.

Current approach:

- CVD: coverage trajectories defined in `04_define_interventions.R` (quadratic fit to NCD-RisC historical data)
- Cancer: screening coverage targets defined in scenario YAML/Excel inputs

What a dedicated module could add:

- Treatment adherence modeling
- Health workforce constraints
- Supply chain / medicine availability
- Facility-level capacity modeling

Comorbidity and Competing Risks

Not Modeled in v1.0

CVD and cancer modules assume conditional independence. No comorbidity interaction or formal competing-risk adjustment is implemented.

Implication:

- Deaths averted from one disease do not reduce the at-risk population for the other
- Summing CVD + cancer results may overestimate total impact
- Acceptable for single-disease policy evaluation; requires adjustment for combined NCD investment cases

Planned for v2.0: Shared population with competing-risk mortality across disease modules.

Part 5 — Closing

Summary

Summary

- Population-based discrete-time Markov framework for Indonesia NCD policy evaluation (2025–2050)
- CVD module: 4 causes, 4 intervention families, calibrated to GBD 2023
- Cancer module: 16 types in 3 tiers, screening interventions for Tier 3
- Economic evaluation via VSL/VSLY with income elasticity scenarios
- Agentic coding approach accelerates development while preserving auditability

Next Steps and Pending Work

Near-term:

- Finalize cancer intervention scenario definitions
- Generate publication-ready output figures for all modules
- Add `renv.lock` for dependency management
- Populate `docs/diagrams/` with model workflow visualizations

Medium-term:

- Implement shared demography module
- Build formal integration layer (unified scenario config, joint output aggregation)
- Add competing-risk adjustment for combined NCD estimates

Longer-term:

- Probabilistic sensitivity analysis (PSA)
- Extend to additional NCD domains (diabetes, COPD)
- Country adaptation toolkit for other LMICs

Key References

Core methodology and data:

- Pickersgill et al. (2022) — Global blood pressure target modeling
- Kontis et al. (2019) — Public health interventions and lives saved
- Robinson et al. (2019) — VSL methodology for global benefit-cost analysis
- Watkins et al. (2025) — Single-pill combination therapies impact

Data sources:

- Chong et al. (2024) — Global CVD burden projections
- Filippini et al. (2021) — Sodium–blood pressure dose-response
- Ettehad et al. (2016) — Blood pressure lowering and CVD outcomes

Thank You

Questions and Discussion

RTSL / University of Washington
Disease Control Priorities
`github.com/Disease-Control-Priorities`

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